Karthikeya Mahenderkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.01
NARAYANA JUNIOR COLLEGE	Class XII (TSBIE)	2022	96.20%
JAYA HIGH SCHOOL	Class X (TSBSE)	2020	10.00%

PROJECTS

ProductPulse AI | AI-Powered Product Analytics & Decision Intelligence Platform

- **Built** an end-to-end product analytics platform processing **200K+ events, 20K orders, and 5K users**, delivering **KPI, funnel, cohort, segmentation, anomaly, and experimentation analytics** through a **React + Spring Boot + PostgreSQL** stack.
- **Developed** a grounded **AI Product Analyst** that converts natural-language questions into **validated SELECT-only SQL**, executes against live database evidence using a **read-only PostgreSQL role**, and generates LLM explanations with **single-statement and query-limit safeguards**.
- **Engineered** statistically rigorous experimentation and anomaly detection using **two-proportion z-tests, Welch's t-tests, chi-square SRM tests, and rolling z-scores**, with safeguards for **underpowered experiments and metric trade-offs** to improve decision reliability.
- **Optimized** read-heavy analytics by precomputing daily metrics into a compact **~1,800-row aggregate table** instead of repeatedly scanning **2M+ events**, while using **indexed live SQL and JdbcTemplate** for dynamic funnel, cohort, and segment analysis.

Real-Time Stock Metrics Tracker

- **Built** a real-time market-monitoring platform using **React, TypeScript, Prisma, and SQLite** to ingest, persist, and visualize live and historical stock data through an interactive analytics dashboard.
- **Engineered** a background market-ingestion pipeline and **rule-based event detection engine** for **52-week high/low breakouts and abnormal-volume spikes**, supporting crossing-based and once-per-day triggers for automated market surveillance.
- **Designed** a pluggable market-data architecture integrating the **Twelve Data API**, mock providers, and **Zod schema validation**, enabling provider-independent testing, reliable data contracts, and extensible market-data ingestion.
- **Implemented REST APIs, Server-Sent Events (SSE), and an event-bus architecture** to stream live market events and analytics, powering **KPI tracking, sector analysis, filtering, historical-date analysis, and intraday charts**.

Event Ticketing Platform

- **Engineered** a full-stack event-ticketing platform using **Spring Boot REST APIs** and a layered architecture for event discovery, ticket purchasing, QR-ticket retrieval, and validation across organizer, staff, and attendee workflows.
- **Prevented ticket overselling** under **concurrent purchases** by combining **PostgreSQL pessimistic locking** with transactional ticket-inventory updates, ensuring consistent allocation when multiple requests target the same event inventory.
- **Implemented role-based access control** with **Keycloak, JWT authorization, and token-claim-based user provisioning**, enforcing distinct **ORGANIZER and STAFF** permissions across event management and ticket-validation operations.
- **Built QR-based ticket verification** with **ZXing**, supporting QR/manual validation and **duplicate-entry prevention**, integrated with a **React/Vite dashboard** for staff to perform fast ticket checks.

SKILLS

Programming Languages: Java, Python

Backend & Frameworks: Spring Boot, Spring Data JPA, Hibernate, REST APIs

Frontend: HTML, CSS

AI Development Tools: Claude Code, GitHub Copilot, Gemini CLI

Libraries & Databases: NumPy, Pandas, Matplotlib, SQL, DBMS

Tools & Build: Git, GitHub, Maven, Jupyter Notebook, Prisma

Soft Skills: Leadership, Team Management, Communication, Decision-Making, Adaptability

RELEVANT COURSES

Computer Science & Programming: Introduction to Programming (C and Data Structures), Fundamentals of Scientific Computing (NumPy, Pandas, Matplotlib), Computational Methods in Materials Science (Python Programming)

Mathematics: Calculus I, Calculus II, Differential Equations, Elementary Linear Algebra

POSITIONS OF RESPONSIBILITY

Publicity Coordinator - Elan & nVision

Led promotional campaigns and social-media outreach for IIT Hyderabad's annual techno-cultural fest, coordinating with design and media teams to drive student engagement and event visibility.

Class Representative

Represented the class in academic coordination, serving as the communication link between students and faculty and addressing student concerns throughout the semester.

EXTRACURRICULAR

- Active Participant in Carrom and Billiards at Milan and Diesta Events
- Volunteered in prayas events and 13th convocation of IIT Hyderabad